



auk 101

An introduction to filtering eBird data

Introducción al filtrado de datos de eBird

Accessing eBird data

Acceso a datos eBird

- The world's birds at your fingertips!
¡Los pájaros del mundo al alcance de tus dedos!



Razorbill by Christoph Moning / Macaulay Library

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- *ebirdst* R package

- eBird Status & Trends products

Paquete R para descargando productos eBird

Status & Trends



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*Paquete R para descargando productos eBird
Status & Trends*

- eBird Basic Dataset (EBD)

- Full eBird database as a text file

*Base de datos completa de eBird como
archivo de texto*



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Download eBird Data Products

eBird provides open data access in several formats to logged-in users, ranging from raw data to processed datasets geared toward more rigorous scientific modeling. For those doing scientific work, please see our more detailed information on [using eBird for science](#).

eBird Basic Dataset (EBD)

The EBD is the core dataset for accessing all raw eBird observations and associated metadata. The EBD is updated monthly (15th of each month), and is [available by direct download through eBird](#) to any logged-in user after completion of a data request form. The data request form allows us to gain some understanding of how the data will be used. Requests are typically approved within 7 days. Data are provided with documentation in spreadsheet format, which can be read by a variety of programs. Although Excel or similar programs work for basic analyses, for larger datasets (>1 million rows) or more sophisticated analyses, we recommend using programs like [R](#). There are several R packages available for summarizing data, including one that is managed here at the Cornell Lab specifically for working with the EBD dataset:

[auk: eBird Data Extraction and Processing with AWK](#)

If you use eBird data in a way that results in a specific conservation action or peer-reviewed publication, please [let us know](#).

[Download the EBD here](#)

<https://science.ebird.org/en/use-ebird-data/download-ebird-data-products>

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Basic Dataset (EBD)

Updated monthly (15th of each month)

Current Version EBD_relMar-2026 – Includes eBird data through Mar-2026

Prepackaged Options

World All species, locations, and dates	39+ GB .tar
Sampling event data Effort data only	5.5 GB .tar

Please note these files are large and might take a long time to download. Files are delivered in .tar format.

Custom Download

Build a custom dataset using these options. Your request will be added to our download queue in the order it was received. We will send you an email when your download is ready. Files are delivered in .zip or .tar format, depending on size. Some species are considered [Sensitive](#) in eBird and are not included in these downloads. Please [contact us directly](#) for access to Sensitive Species data.

Species: ☐ All species

☒ Choose species...

Region: ☒ All regions

☐ Choose region...

Date Range: ☒ All dates

☐ Choose date range...

Options: ☐ Include sampling event data - *Recommended* [?](#)

☐ Include unvetted data [?](#)

[Submit Request](#)

<https://science.ebird.org/en/use-ebird-data/download-ebird-data-products>

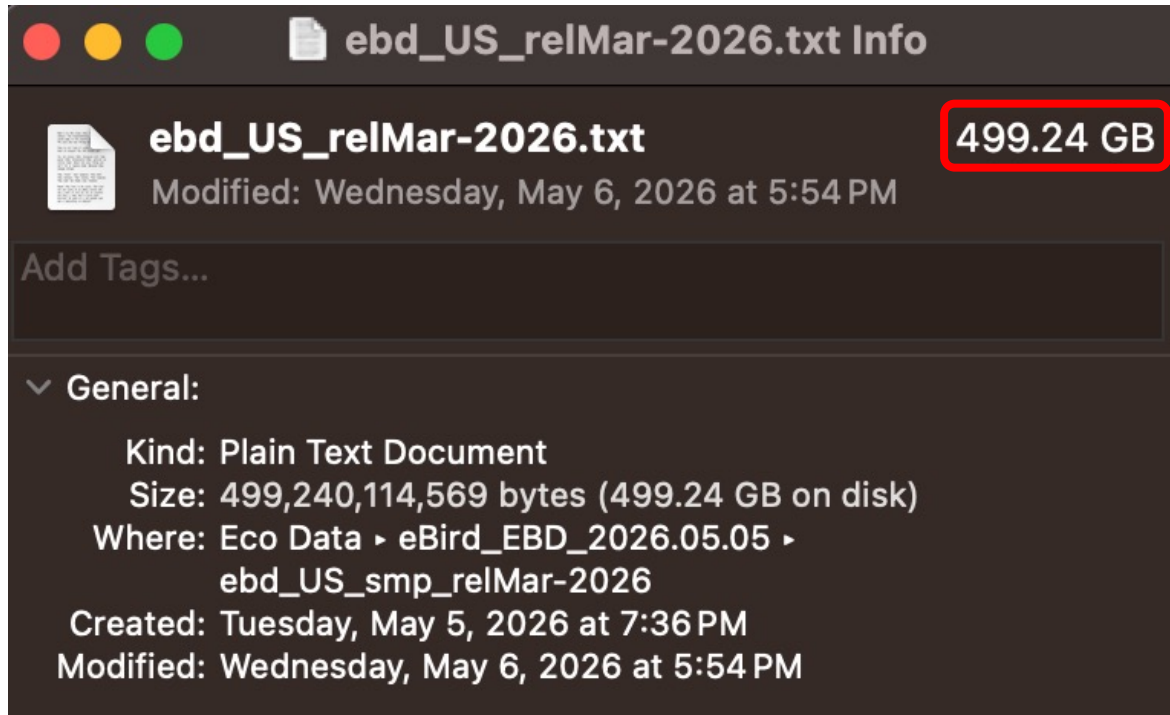


cccliff1126 / Pixabay

But wait, there's a problem!
The EBD file is HUGE!



*¡Pero espera, hay un problema!
¡El archivo EBD es ENORME!*





Can't feasibly filter EBD file in Excel or R

No se puede filtrar un archivo EBD en Excel o R

Extremely time-intensive, assuming file even opens!

¡Extremadamente laborioso en tiempo, suponiendo que siquiera se abra el archivo!





Somehow...

De alguna manera...

Managing file size

Manejando el tamaño de archivos

- Possible to filter file contents before import to R

Es posible filtrar el contenido de archivos antes de importar a R



Razorbill by Lukasz Ifczok / Macaulay Library

Managing file size

Manejando el tamaño de archivos

- Possible to filter file contents before import to R

Es posible filtrar el contenido de archivos antes de importar a R

- Method uses “**AWK**”

- Unix utility/programming language for processing text data

*El método usa “**AWK**” un lenguaje de programación Unix para procesar datos de texto*



Razorbill by Lukasz Ifczok / Macaulay Library

Who here has used **AWK**?
*¿Quien ha usado **AWK**?*

Who here has used **AWK**?

*¿Quien ha usado **AWK**?*

...

Has anyone even heard of **AWK**?

*¿Alguien ha oido sobre **AWK**?*

```
ebd_relFeb-2018 — -bash — 116x28
ag-clo-mes335:ebd_relFeb-2018 mes335$ cd /Users/mes335/data/ebird/ebd_relFeb-2018
ag-clo-mes335:ebd_relFeb-2018 mes335$ ls
awk
ebd_relFeb-2018_SG.txt
ebd_relFeb-2018_clean.txt
ebd_relFeb-2018_rollup.txt
ebd_relFeb-2018_subset.txt
ebd_sampling_relFeb-2018_SG.txt
ebd_sampling_relFeb-2018_clean.txt
ebird_country-state.txt
ag-clo-mes335:ebd_relFeb-2018 mes335$ head -20 ebd_relFeb-2018_clean.txt | awk -F '\t' '{print $25,$26,$27,$5,$9}'
LATITUDE LONGITUDE OBSERVATION DATE COMMON NAME OBSERVATION COUNT
35.1085767 -120.6265268 2008-11-05 Belted Kingfisher 1
35.1085767 -120.6265268 2008-11-05 Wilson's Snipe 1
35.1085767 -120.6265268 2008-11-05 Ruby-crowned Kinglet 15
35.1085767 -120.6265268 2008-11-05 California Gull 35
35.1085767 -120.6265268 2008-11-05 Mallard 40
35.1085767 -120.6265268 2008-11-05 Bewick's Wren 1
8.52111 -83.39991 2004-06-18 Rufous Piha X
8.52111 -83.39991 2004-06-18 Thick-billed Euphonia X
46.3024735 -91.0284233 2008-08-16 Chestnut-sided Warbler 4
46.3024735 -91.0284233 2008-08-16 American Robin 9
33.7968382 -117.7532673 1971-01-02 Spotted Towhee 4
33.7968382 -117.7532673 1971-01-02 Red-tailed Hawk 5
33.7968382 -117.7532673 1971-01-02 Anna's Hummingbird 11
33.7968382 -117.7532673 1971-01-02 Bewick's Wren 5
37.1434192 -114.0232372 2008-10-25 Spotted Towhee 6
43.1812545 -88.0415756 2008-08-06 Mallard 5
43.1812545 -88.0415756 2008-08-06 Red-winged Blackbird 3
41.059032 -112.238045 2006-08-13 Black-necked Stilt 1
41.059032 -112.238045 2006-08-13 California Gull 1
ag-clo-mes335:ebd_relFeb-2018 mes335$
```

AWK 🤔

Terminal



```
awk-example.txt x
1
2 BEGIN {
3   FS = OFS = " "
4   split("Cyanocitta cristata Perisoreus canadensis", speciesValues, " ")
5   for (i in speciesValues) species[speciesValues[i]] = 1
6   split("CA US", countryValues, " ")
7   for (i in countryValues) countries[countryValues[i]] = 1
8 }
9 {
10  keep = 1
11  if (keep == 1 && ($6 in species)) {
12    keep = 1
13  } else {
14    keep = 0
15  }
16  if (keep == 1 && ($14 in countries)) {
17    keep = 1
18  } else {
19    keep = 0
20  }
21  if (keep == 1 && ($26 >= -100 && $26 <= -80 && $25 >= 37 && $25 <= 52)) {
22    keep = 1
23  } else {
24    keep = 0
25  }
26  if (keep == 1 && ($27 >= "2012-01-01" && $27 <= "2012-12-31")) {
27    keep = 1
28  } else {
29    keep = 0
30  }
31  if (keep == 1 && ($28 >= "06:00" && $28 <= "09:00")) {
32    keep = 1
33  }
34}
```

Downsides of AWK

Desventajas de AWK

01 Must learn a new programming language

Debo aprender un nuevo lenguaje de programación

02 Only accessible via command line

Solo accesible por línea de comandos

03 Part of workflow occurs outside of R

Parte del flujo de trabajo ocurre fuera de R

auk R package

auk 0.3.3

Vignette

Function Reference

auk: eBird Data Extraction and Processing in R

License GPL v3

Mac OSX & Linux passing

Windows passing

coverage 79%

R Peer Reviewed

CRAN 0.3.2

downloads 13K



Overview

eBird is an online tool for recording bird observations. Since its inception, over 600 million records of bird sightings (i.e. combinations of location, date, time, and bird species) have been collected, making eBird one of the largest citizen science projects in history and an extremely valuable resource for bird research and conservation. The full eBird database is packaged as a text file and available for download as the [eBird Basic Dataset \(EBD\)](#). Due to the large size of this dataset, it must be filtered to a smaller subset of desired observations before reading into R. This filtering is most efficiently done using AWK, a Unix utility and programming language for processing column formatted text data. This package acts as a front end for AWK, allowing users to filter eBird data before import into R.

Installation

```
# cran release
install.packages("auk")

# or install the development version from github
# install.packages("remotes")
remotes::install_github("CornellLabofOrnithology/auk")
```

auk requires the Unix utility AWK, which is available on most Linux and Mac OS X machines. Windows users will first need to install [Cygwin](#) before using this package. Note that **Cygwin must be installed in the default location** (C:/cygwin/bin/gawk.exe or C:/cygwin64/bin/gawk.exe) in order for auk to work.

Vignette

Full details on using auk to produce both presence-only and presence-absence data are outlined in the vignette, which can be accessed with `vignette("auk")`.

Best Practices website

eBird Best Practices

Welcome

1 Introduction and Setup

2 eBird Data

3 Habitat Covariates

4 Modeling Encounter Rate

5 Modeling Occupancy

6 Modeling Relative Abundance

References

Best Practices for Using eBird Data

Matthew Strimas-Mackey, Alison Johnston, Wesley M. Hochachka, Viviana Ruiz-Gutierrez, Orin J. Robinson, Elliot T. Miller, Tom Auer, Steve Kelling, Daniel Fink

2019-04-18

Welcome

This book is currently under active development. Please explore and give us feedback, but be aware that some sections are incomplete and others may still contain bugs.

Best Practices for Using eBird Data is a supplement to *Best practices for making reliable inferences from citizen science data: case study using eBird to estimate species distributions* (Johnston et al. 2019). This paper describes the challenges associated with making inferences from biological citizen science data and proposes a set of best practices for making reliable estimates of species distributions from these data. Throughout, the paper uses eBird, the world's largest biological citizen science project, as a case study to illustrate the good practices. This book acts as a supplement to the paper, showing readers how to implement these best practices within R using real data from eBird. After completing this book, readers should be able to extract data from the eBird database suitable for their own studies, process these data to prepare them for robust analyses, collect environmental covariates for modeling, and fit and assess models estimating encounter rate, occupancy, and relative abundance. Readers should be comfortable with the R programming language, and read the [Prerequisites](#) and [Setup](#) sections of the introduction, before diving into this book.

A preprint of the paper associated with this book is currently [available on bioRxiv](#).

Who created auk?

¿Quienes crearon auk?

Matthew Strimas-Mackey (author, maintainer)

Eliot Miller (author)

Wesley Hochachka (author)





`auk` acts as a front end for `AWK`
`auk` actúa como interfaz para `AWK`

Allows users to filter eBird data ***before*** import into R
*Permite a los usuarios filtrar datos de eBird ***antes*** de importarlos a R*

auk workflow

auk flujo de trabajo

data access

1

acceder a datos

filter

2

filtrar

import

3

importar

process

4

procesar

prep data for analysis

5

prep datos para analisis

auk workflow

auk *flujo de trabajo*

- 1 data access
acceder a datos
- 2 filter
filtrar
- 3 import
importar
- 4 process
procesar
- 5 prep data for analysis
prep datos para analisis

Important functions

Funciones importantes

1

data access functions

funciones de acceso a datos

2

filtering functions

funciones de filtrado

3

importing functions

funciones de importación

4

processing functions

funciones de procesamiento

5

analysis prep functions

funciones de preparación del análisis

Important functions

Funciones importantes

`auk_ebd()`

References eBird data file

`auk_sampling()`

References eBird sampling event file

1

data access functions

funciones de acceso a datos

2

filtering functions

3

importing functions

4

processing functions

5

analysis prep functions

Important functions

Funciones importantes

Apply filters

`auk_filter()`

Apply filters to eBird file using AWK

Spatial

`auk_bbox()` `auk_extent()`

Filter by spatial bounding box/extent

`auk_bcr()`

Filter by Bird Conservation Region

`auk_country()` `auk_county()` `auk_state()`

Filter by country/county/state

`auk_distance()`

Filter by distance travelled

1

data access functions

2

filtering functions

funciones de filtrado

3

importing functions

4

processing functions

5

analysis prep functions

Important functions

Funciones importantes

Time

`auk_date()` `auk_year()`

Filter by date/year

`auk_duration()` `auk_time()`

Filter by duration/start time

`auk_last_edited()`

Filter by last edited date

Other

`auk_breeding()`

Only include observations w/breeding codes

`auk_complete()`

Remove incomplete checklists

`auk_project()` `auk_protocol()`

Filter by project code/protocol

`auk_species()`

Filter by species

1

data access functions

2

filtering functions

funciones de filtrado

3

importing functions

4

processing functions

5

analysis prep functions

Important functions

Funciones importantes

`read_ebd()` `read_sampling()`

Read an EBD file

`auk_zerofill()` `collapse_zerofill()`

Read and zero-fill an eBird data file

1

data access functions

2

filtering functions

3

importing functions

funciones de importación

4

processing functions

5

analysis prep functions

Important functions

Funciones importantes

`auk_select()`

Select a subset of columns

`auk_split()`

Split an eBird data file by species

`auk_rollup()`

Roll up eBird taxonomy to species

`auk_unique()`

Remove duplicate group checklists

1

data access functions

2

filtering functions

3

importing functions

4

processing functions

funciones de procesamiento

5

analysis prep functions

Important functions

Funciones importantes

`auk_zerofill()` `collapse_zerofill()`

Read and zero-fill an eBird data file

`filter_repeat_visits()`

Filter to repeat visits for hierarchical modeling

`format_unmarked_occu()`

Format for occupancy modeling with unmarked

1

data access functions

2

filtering functions

3

importing functions

4

processing functions

5

analysis prep functions

funciones de preparación del análisis

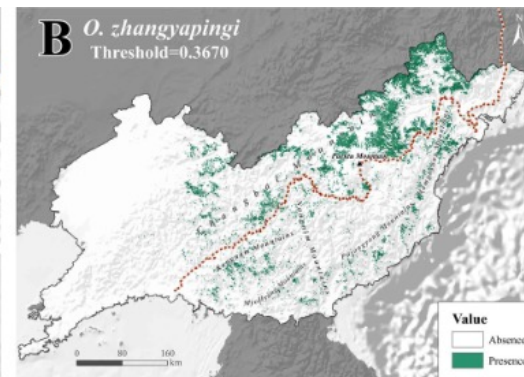
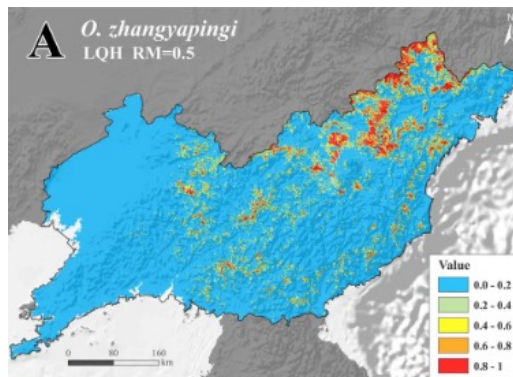
Example uses of filtered data

Ejemplos de usos de datos filtrados

- `auk_zerofill()` `collapse_zerofill()`

- Species distribution & resource selection models

Distribución de especies y modelos de selección de recursos



Peng et al. 2023 / Global Ecology and Conservation

Example uses of filtered data

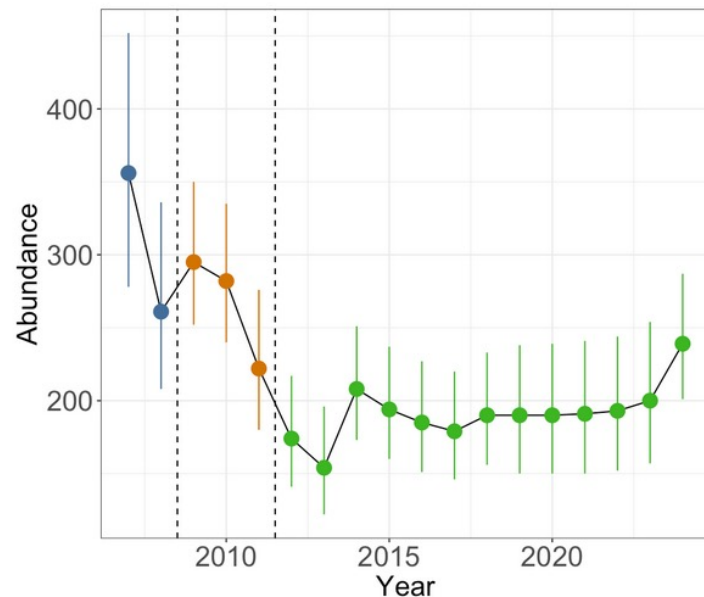
Ejemplos de usos de datos filtrados

- `auk_zerofill()` `collapse_zerofill()`
 - Species distribution & resource selection models

- `filter_repeat_visits()`

- Hierarchical models

Modelos jerárquicos

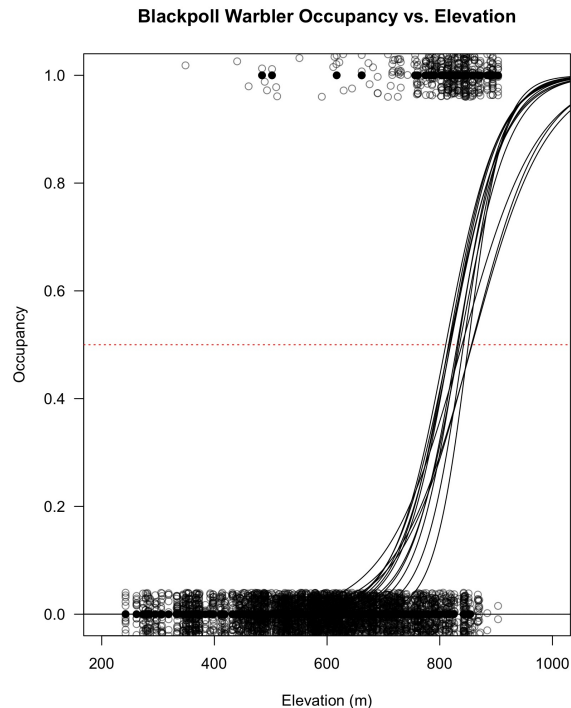


Rushing 2025 / WILD8370 Bayesian Models for Conservation Science

Example uses of filtered data

Ejemplos de usos de datos filtrados

- `auk_zerofill()` `collapse_zerofill()`
 - Species distribution & resource selection models
- `filter_repeat_visits()`
 - Hierarchical models
- `format_unmarked_occu()`
 - Occupancy modeling with *unmarked*
Modelos de ocupación con unmarked



Van Tatenhove et al. 2018 / Hubbard Brook Cooperators' Meeting

Example uses of filtered data

Ejemplos de usos de datos filtrados

- `auk_zerofill()` `collapse_zerofill()`

- Species distribution & resource selection models

Distribución de especies y modelos de selección de recursos

- `filter_repeat_visits()`

- Hierarchical models

- `format_unmarked_occu()`

- Occupancy modeling with *unmarked*

Today's focus

El enfoque de hoy



Questions?

¿Preguntas?

Next up: Downloading data

A continuación: Descarga de datos